1. A hospital needs software designed specifically to manage patient records, appointments and medical histories. Identify the software category that best describes this application.

A) General-purpose software
B) Utility software
C) Bespoke software
D) Open-source software

Answer:

_____

[1]

2. An operating system must guarantee that an industrial robot receives sensor data and responds within a strict deadline. Identify the most suitable type of operating system.

A) Batch processing operating system
B) Real-time operating system (RTOS)
C) Single-user single-tasking operating system
D) Network operating system

[1]

3. Which is a major advantage of a Command-Line Interface (CLI) compared with a Graphical User Interface (GUI)?

A) It always requires more RAM.
B) It supports efficient scripting and automation.
C) It uses icons instead of commands.
D) It requires no knowledge of commands.

[1]

4. Which statement correctly describes the relationship between Artificial Intelligence (AI), Machine Learning (ML) and Deep Learning (DL)?

A) AI is a subset of ML, and ML is a subset of DL.
B) ML is the broadest field, while AI is a subset of ML.
C) AI is the broadest field, ML is a subset of AI, and DL is a subset of ML.
D) AI, ML and DL are unrelated fields.

[1]

5. A smartphone application places virtual arrows and labels over the user's live camera view of a building. This is an example of:

A) Virtual Reality
B) Augmented Reality
C) Batch Processing
D) Natural Language Processing

[1]

6. A programmer describes the required result of a database query without specifying the exact sequence of operations used to obtain it. Identify the programming paradigm.

A) Imperative programming
B) Declarative programming
C) Assembly programming
D) Procedural machine code

[1]

7a. State the primary purpose of a utility program.

_____

_____

[1]

7b. State the primary purpose of a software library.

_____

_____

[1]

7c. Explain what a translator does to source code before it can be executed by the CPU.

_____

_____

[1]

8a. Describe the main difference between a single-user operating system and a multi-user operating system.

_____

_____

[1]

8b. Explain one resource-management or security challenge that can occur in a multi-user operating system.

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_____

_____

_____

[2]

8c. A company collects all payroll transactions during the day and processes them together at midnight without user interaction. Identify the processing mode.

_____

_____

[1]

9a. A network administrator manages remote servers through a slow connection. Justify why a CLI may be more suitable than a GUI.

[1]

9b. A new employee uses a point-of-sale system and has little technical experience. Justify why a GUI may be more suitable.

[1]

9c. Identify one advantage and one disadvantage of using a Voice User Interface (VUI) in an interactive system.

[2]

10. Compare High-Level Languages (HLL) and Low-Level Languages (LLL) in terms of hardware dependence and direct control of memory.

[2]

11a. State one advantage and one disadvantage of using Internal CSS compared with External CSS.

[2]

11b. Write the exact HTML code needed inside the <head> section to connect an external stylesheet named main.css.

[2]

12. Write HTML code to create a web form containing: Full Name (text input); Date of Birth (date input); Email Address (email input); Preferred Course (drop-down list with at least three options); Study Mode (Online and On-campus radio buttons); Personal Statement (textarea); and Clear and Send buttons.

[10]

13. The following SDLC stages are shown: Planning → ? → Development → Testing → Maintenance. Name the missing stage.

[1]

14. A software team is creating an online learning platform. The customer expects requirements to change after each prototype. Identify the more suitable model — Agile or Waterfall — and give one reason.

[2]

15a. Describe one advantage and one disadvantage of using observation rather than questionnaires to collect system requirements.

[2]

15b. Give one example of qualitative data and one example of quantitative data that could be collected when evaluating a new school information system.

_____
_____
_____
_____

[2]

TOTAL: _____ / 40 marks

Mark Scheme — Variant B

No.	Expected answer	Marks	Comments
1	C — Bespoke software	1	Correct option.
2	B — Real-time operating system (RTOS)	1	Correct option.
3	B — It supports efficient scripting and automation.	1	Accept equivalent wording about low overhead or automation.
4	C — AI is the broadest field; ML is a subset of AI; DL is a subset of ML.	1	All relationships must be correct.
5	B — Augmented Reality	1	Correct option.
6	B — Declarative programming	1	Correct paradigm.
7a	Performs maintenance or management tasks such as backup, compression, disk cleanup or virus scanning.	1	Any valid utility-program purpose.
7b	A collection of reusable, pre-written code modules, routines or functions.	1	Must mention reusable code/components.
7c	Converts source code into machine code or another executable/intermediate form that can be processed by the computer.	1	Accept a technically valid compiler/interpreter explanation.
8a	Single-user supports one user at a time; multi-user supports multiple users accessing system resources.	1	Clear contrast required.
8b	Examples: fair CPU/RAM allocation; preventing users from accessing or overwriting other users' data; separate permissions or sessions.	2	1 mark for challenge; 1 mark for explanation.
8c	Batch processing.	1	Correct processing mode.
9a	CLI transfers mainly text, uses little bandwidth and supports commands/scripts for remote administration.	1	Any one valid justification.
9b	GUI uses icons, menus and buttons, making it intuitive and requiring little command syntax knowledge.	1	Any one valid justification.
9c	Advantage: hands-free, accessible or natural interaction. Disadvantage: noise, accents, privacy concerns or recognition errors may reduce accuracy.	2	1 mark each.
10	HLL is generally portable and hardware-independent; LLL is tied to a processor architecture. LLL gives more direct control of registers and memory addresses; HLL abstracts memory management.	2	1 mark for hardware dependence; 1 mark for memory control.
11a	Advantage: useful for page-specific styles and no separate file is needed. Disadvantage: styles are harder to reuse and maintain across many pages.	2	1 mark each.
11b	<link rel="stylesheet" href="main.css">	2	1 mark for link element; 1 mark for correct attributes.
12	Correct HTML form with all seven required elements: text, date, email, select list, two same-name radio buttons, textarea, reset/clear and submit/send buttons.	10	Award marks for each required element and correct structure.
13	Analysis / Requirements analysis / System analysis	1	Any equivalent valid stage name.
14	Agile. Requirements can change between iterations, and feedback from prototypes can be incorporated during development.	2	1 mark for Agile; 1 mark for reason.
15a	Advantage: observes real behaviour and actual work	2	1 mark each.

	processes. Disadvantage: time-consuming, observer effect, or activities may not occur during observation.		
15b	Qualitative: user comments, opinions or descriptions of usability. Quantitative: number of errors, response time, rating out of 5 or number of users.	2	1 mark for each valid example.

TOTAL: 40 marks